



Exploring the Impact of Artificial Intelligence on Research Ethics - A Systematic Review

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Abstract

The rapid advancement of artificial intelligence (AI) has profoundly transformed many people's lives, ChatGPT being a clear example, whose capabilities have substantially influenced the automation of tasks such as writing texts and providing information sources for researchers. This review article aims to understand the impact of AI on academic writing and why its use can be considered plagiarism. The Prism method was used to analyze the studies, which initially totaled 824, and after excluding them for duplicity and by title, a total of 137 were left, of which we proceeded to review those that were open access and those that were related to the study, obtaining a total of 54 manuscripts closely related to the research topic; the results were then segmented into three questions, What are the text-matching tools that help to identify the use of AI in article writing?, What strategies are used to regulate the use of AI in writing scientific articles? and What techniques are used to detect AI-generated text?, which were key to comparing the results of the review with the findings of the authors belonging to the literature review. The results showed that, from 2022 onwards, AI became a recurring topic among researchers in China and the United States, allowing the emergence of popular techniques and software such as Turnitin or GPTZero to identify when these linguistic models were used in writing tasks, as well as strategies to regulate their use in controlled environments. In the end, it was concluded that there is a fine line between ethics and abuse of AI capabilities, and further research and study of different techniques is recommended to recognize when these tools are blatantly used.

Keywords Artificial intelligence · Plagiarism · Ethics · Article · Writing

Introduction

As new topics and phenomena emerge that attract the attention of researchers and, consequently, become the focus of new research, plagiarism is one of the most serious violations of standards that can arise (Subedi et al., 2024); and that's why, at the same time, text-matching tools appear that facilitate the detection of plagiarism in this type of work

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(El-Rashidy et al., 2024). While Turnitin is a plagiarism detector par excellence founded in 1999 and gained credibility over the years, other tools such as Copyscape, created in 2004, or Academia, introduced in 2018, also meant a great pillar for the fight against plagiarism (Foltýnek et al., 2020). Another tool that was introduced to the public in 1999 was Plagiarism.org, which allowed academics, journal editors, and even students to compare their work with thousands of other available on the internet (Dalton, 1999).

Recently, AI-powered products have taken their place in changing how we conduct our day-to-day activities (Saygın & Kabakçı, 2023), with OpenAI's sophisticated chatbot, ChatGPT, being a relevant example of these functionalities, demonstrating how its functionalities greatly influence when writing text or generating information sources for researchers (H. Jo & Park, 2024). The versatility of this model makes it a tool that can assist in research writing, from formulating hypotheses to creating convincing results and writing publication-ready manuscripts (Elbadawi et al., 2024). However, its ability to reference the literature is still far from replicating human judgement (Fleckenstein et al., 2024).

Given the above, academic writers may incorporate AI-generated text into their work, which threatens academic integrity (Miao et al., 2024). The excellent feedback quality of AI tools has driven plagiarism among students when using AI-generated text (Sysoyev, 2024). This points to the urgency of re-evaluating some traditional ethical norms in research (Kasani et al., 2024), as well as remembering that the work of ChatGPT cannot replace human judgement, and any results generated must be peer-reviewed before application in critical decisions (Salvagno et al., 2023) and subsequent publication in relevant media, such as academic writing and dissemination of scientific knowledge (Peres, 2024).

With all of the above, the reason for developing this work lies in the need to recognize and establish ethical values when using text generation tools in academic writing articles, due to the high difficulty of identifying when these tools were used. One study showed that when using "GPT-2 Output Detector", a tool that identifies the use of AI in writing, in test texts, 99.98% were identified as "false", while the remaining 0.02% remained true to the original human-written summaries (Gao et al., 2023).

In this review, we seek to understand the impact of AI on academic writing and to what extent its use is considered plagiarism. Therefore, the aim of this paper is to explore cases where generative chatbots violate ethical issues, as well as to develop some recommendations when using them and avoid falling into plagiarism, as the inaccurate information generated can have repercussions in both medical science and engineering.

Throughout the review, there will be an exploration and support of the methodology used to collect relevant information according to the development of the research, the results obtained after the analysis of the data collected, opening a critical debate on the results obtained, the conclusions drawn from the information analyzed and finally, the references used for the argument and support of the article.

Literature review

Impact of AI

Clearly, the use of AI for academic purposes is becoming increasingly common. For example, research (Titko et al., 2023) explored academic staff perceptions of AI-based text generators. To achieve the objective, they conducted a questionnaire distributed across institutions in 10 European countries, resulting in a lack of skills in the area of AI on the part of

academic staff. They found a lack of knowledge about AI among teachers and the need for training and regular reviews to avoid plagiarism. On the other hand, the study (E. Jo & Kim, 2023) presents a new approach to poetry writing using AI with high school students. They conducted an experiment where 148 high school students were asked to write poetry using AI, asking the AI to opt for a first-person stance. They found that AI can be useful in the "expression and rewriting" stages, concluding that students must have a solid knowledge of the subject. Also, the paper (Črček & Patekar, 2023) explored the ways in which students use the ChatGPT tool for writing tasks and ethical perspectives on its use. They interviewed 201 students about using this AI and found that more than half used it for writing tasks, but some used it to generate ideas, summaries, paraphrase and write sections of text. They conclude that universities should provide clear guidelines to promote the ethical use of AI-based chatbots and emerging technologies.

Impact of AI and Research Ethics

In recent years, research and reviews have been conducted that contribute to the scientific construction of the use of AI. For example, (Singh, 2023) analyses the impact of ChatGPT on plagiarism and university academic writing. They used a case study methodology consisting of interviews with three teachers through semi-structured questioning on biographical questions and academic logs. They found that AI can be useful for writing, but it requires responsible use by students and an educational environment that promotes its appropriate use. Similarly, the study (Kutbi et al., 2024) proposed developing a machine learning-based model to identify inconsistency in student-written essays and ChatGPT-generated texts. They used Inverse Document Frequency (TF-IDF) based data processing and transformation over random sampling and logistic regression to identify inconsistency in AI-generated text, resulting in a familiarity accuracy of 98%. They conclude and put much emphasis on suggesting measures to explore a good application of language models to identify machine-generated text and avoid academic inconsistency. In addition, the study (Guleria et al., 2023) took care to highlight the use of assisted AI and chatbots such as ChatGPT in scientific writing and research. To do this, they asked the AI to write an article from scratch and then run it through GPTZero, identifying that, although the content generated was correct, it lacked credibility in the references, as they were created by the AI and could not be found on the web. They conclude that this technology has positive and negative impacts in different fields and recommend policies that regulate its use in content generation.

Research Ethics

In this section, the paper (Lund et al., 2023) identifies the most significant ethical dilemmas when using ChatGPT, such as copyright and citation of references. They conducted extensive research on the use of AI in writing professional articles, finding that they lacked credibility when properly citing references used in the text. They conclude that academia and research institutions should promote creative measures and innovative ideas when commissioning work in this area. Also, the implications of language models in publishing have been increasing, and research by (Ring, 2023) examined the implications of language models in scientific publishing. They proposed to ask that model directly what it thought about this issue, having a neutral point of view, because although the text was artificially generated, it did not lack truth in its arguments. However, it cannot reference

the information generated. They identified that, although AI can generate texts with neutral arguments, it cannot reference information, which generates ambivalence about its use in scientific publication. If we focus on writing and writing, the study (Abani et al., 2023) presents an investigation to determine whether ChatGPT can generate scientific articles convincing to human judgment. To do so, they selected three research articles to construct an abstract with the language model, which was reviewed by different text-matching tools that facilitate plagiarism detection and AI identification, showing texts that were too wordy, superficial and lacking original creativity. They concluded that these language models can be considered a threat to scientific integrity but suggest a good use of these tools among students in university programs. In times of pandemic, a great growth in AI development was identified. For example, the study (Balel, 2023) analyses the rise of AI in the context of COVID-19. Based on the exhaustive investigation of the database used by ChatGPT, they identified that the data generated comes from academic sources and any web page, completely ignoring what information it will use for text generation. They concluded that language models such as ChatGPT do not guarantee academic quality, and care should be taken against plagiarism. However, there is also potential to explore new possibilities with AI.

Materials and Methods

Regarding the development of the methodology, this section must be structured into elements that explore the purpose of the systematic review, the research questions, the type of study employed, the search strategy in different databases, and the application of inclusion and exclusion criteria.

Purpose and Research Questions

Based on the literature review, this review seeks to explore the impact of AI on different ethical aspects, specifically in the field of research and writing texts generated by these tools. The research questions are described below:

- RQ1: What are the text-matching tools to identify the use of AI in article writing?
- RQ2: What strategies are used to reduce the use of AI in writing scientific articles?
- RQ3: What techniques are used to detect AI-generated texts?

Type of Study

The type of methodology selected is the Systematic Literature Review (SLR), which provides a methodical process for collecting and analyzing existing evidence to answer specific questions, assess the validity of findings, synthesize knowledge in each field, identify future research topics and evaluate theories of why certain phenomena occur (Barrantes-Saucedo et al., 2023), (Page, McKenzie, et al., 2021). The PRISMA methodology, which has the advantage of going through the different stages of the review, shows the number of records identified, as well as the criteria used for their exclusion and inclusion (PRISMA, 2020). This methodology was selected for its focus on transparency and completeness in constructing systematic literature reviews, facilitating

decision making, assessing reliability and analyzing findings, providing greater clarity in the results obtained and ensuring the relevance of the review in the scientific field (Page, Moher, et al., 2021).

Search Strategy

The data collection for the development of the review was based on an exhaustive analysis of various databases, such as Scopus, EBSCO Host, IEEE Xplore and others. Initially, 815 studies were identified, 9 of which were excluded due to duplicity; finally, 54 studies were selected by applying the inclusion and exclusion criteria structured by the PRISMA methodology. The search string was composed of the keywords and related terms that encompass the review; once prepared, it is applied in the different databases, varying the filters for each of them: ("Artificial intelligence" OR AI OR chatbot OR ChatGPT) AND (plagiarism OR ethic OR ethical OR ethics) AND (strategies OR methods OR detection OR identify OR recognize) AND (article OR publication OR publishing OR "higher education" OR education OR writing OR research OR redaction).

Inclusion and Exclusion Criteria

The inclusion and exclusion criteria that were applied in this systematic review are detailed in Table 1 below.

Identification and screening processes were applied to collect studies in each database, finally counting the total number of studies included in the review. As a first point, 824 studies were found, leaving only 815 after eliminating nine duplicates. Next, 602 studies were excluded by title and 76 by abstract because they were not related to the central theme of the review, leaving a total of 137 studies. Next, a full-text availability check was performed, leaving six more studies and updating the total count to 131 studies. Finally, a full-text review of the remaining studies was carried out, with 77 excluded for not answering the PICO questions correctly, thus including 54 studies in the systematic review, as presented in Fig. 1.

Table 1 Inclusion and exclusion criteria

Inclusion	Exclusion
I01: Manuscripts from 2019 to 2024	E01: Manuscripts written in a language other than English
I02: Manuscripts must be written in the English language	E02: Manuscripts do not answer the research questions (PICO)
I03: Original manuscripts, conference papers or books	E03: Manuscripts do not relate to the research topic
I04: Publicly accessible manuscripts	
I05: Manuscripts with full PDF document	



This section analyses the findings found in the different databases, indicating some key criteria such as the types of studies, the approach used, and the number of studies found in each database. Figure 2 describes the bibliometric analysis of the most recurrent keywords encompassing the use of AI in writing.

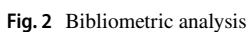


Figure 2 presents the main topics related to AI, which are the major focuses of study and analysis by different authors, highlighting those that receive the most attention and co-occurrence in research. The ramifications that stand out the most are topics that address crucial issues such as the application of AI in education, ethical considerations inherent to its development and use, implementation in various sectors, automated learning, chat-bots and issues associated with plagiarism. There is also a strong link to discussions on scientific writing, privacy and the development of methods to detect its use in professional publications.

Next, Fig. 3 considers the document type found in the different databases. This shows the visual distribution that divides the types of documents used in this review into three sections, each belonging proportionally to the percentage corresponding to the type of study used. The largest section, comprising 87% of the total, comprises original articles, indicating a high frequency of use of this source type in the publications consulted from the repositories. In second place are conference papers, which represent 11% of the total, and finally, the smallest section, corresponding to 2%, comprises books, whose category had minimal participation in the analysis of the central theme addressed in the review but of equal importance in complementing the knowledge identified.

Figure 4 plots the approach used by each study collected from each database. In this figure, we can visualize the distribution that classifies the documents found according to the type of methodology and the database from which they were extracted, allowing us to identify patterns and trends regarding the predominant approaches in the review. The SCOPUS database and studies found in other databases host the largest number of documents with a quantitative approach, followed by other databases such as EBSCO and IEEE, although to a significantly lesser extent. On the other hand, when analyzing the documents with a qualitative focus, it is observed that SCOPUS, with other databases, exhibits a greater amount of this type of research and, again, is followed to a lesser extent by EBSCO and IEEE. Finally, a third methodology combining both approaches is present in the diagram, but with a lower presence than the first two approaches.

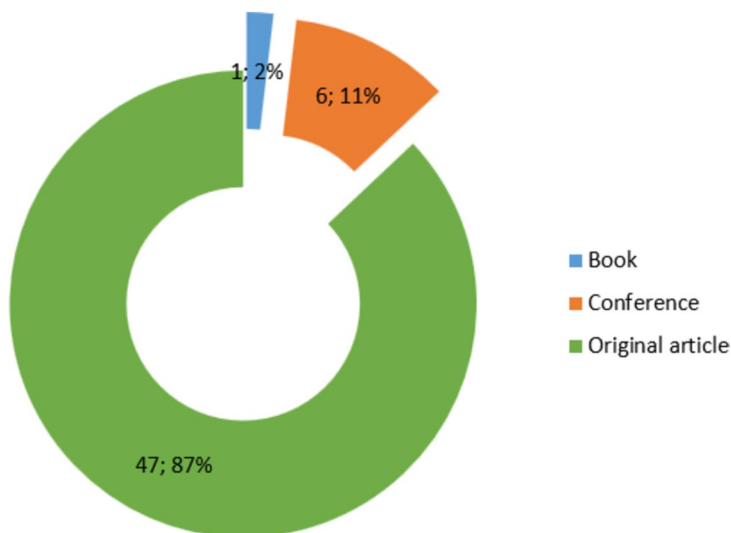


Fig. 3 Type of document

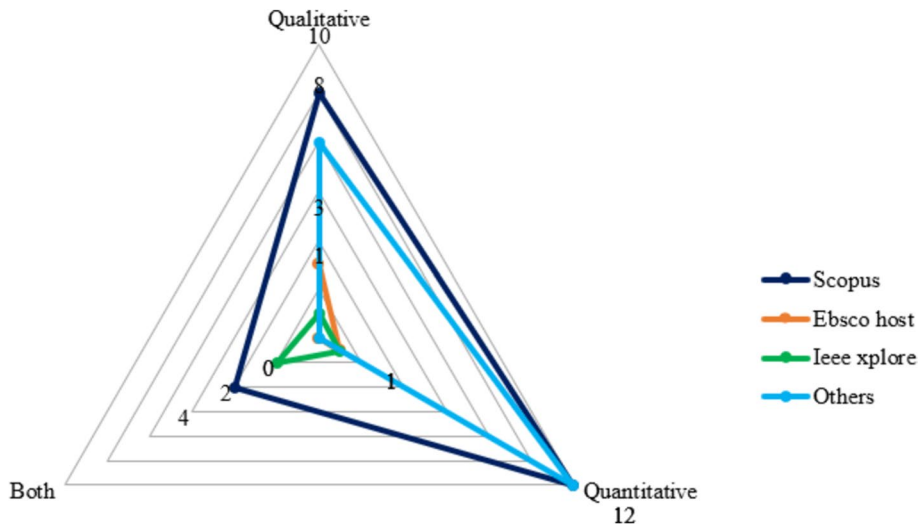


Fig. 4 Study by approach

Figure 5 below shows the number of studies found in each database, which were included in the final version of this review. A bar chart of the number of documents obtained in different databases is presented, which is very useful for understanding the main sources consulted and their relative importance within the review. The SCOPUS database was the most used, contributing 26 documents, highlighting its relevance as a primary source of information for the central theme. The same can be said of other grouped databases, which contributed 20 documents, thus enriching the systematic review. On the other hand, EBSCO and IEEE presented a smaller, but no less important

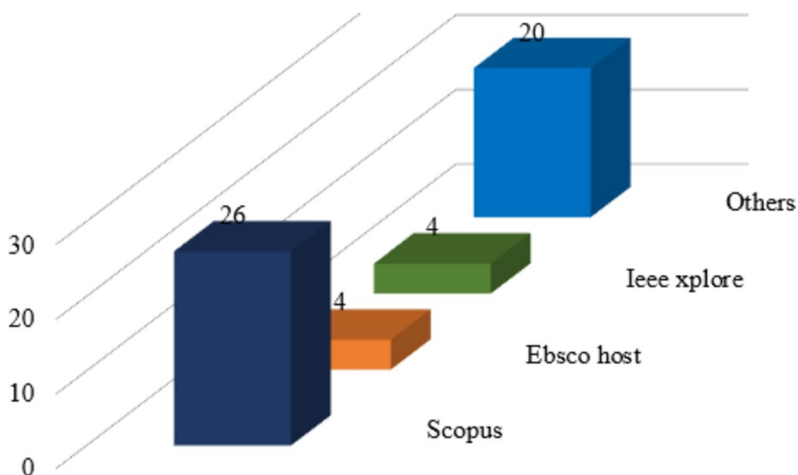


Fig. 5 Number of studies per database

contribution, contributing four studies each, highlighting the importance of consulting various sources to obtain a more complete view of the research topic.

Finally, Fig. 6 describes the frequency of publication of these studies from 2019 to 2024 and the place of publication. It is possible to visualize that from 2019 to 2021, the topic of AI and its use in writing was not much touched upon, as it was not until 2022 that the first version of ChatGPT was released to the public, starting to have some relevant studies. It was not until 2023 onwards that it became a recurring theme among researchers, with the highest frequency of publication in China and the United States, countries where AI had a significant impact and is currently the focus of various conversations.

Discussion

In the method chapter, the questions underpinning this review's development were briefly presented, indicating key aspects such as techniques for identifying the use of AI in writing and what strategies to put in place to regulate its use. Once the selected manuscripts have been analyzed, these research questions are answered and substantiated.

RQ1 What are the text-matching tools that help to identify the use of AI in article writing?

Several tools can be used to detect plagiarism attempts when using language models, such as ChatGPT when generating text in professional writing.

The review results show Turnitin, GPTZero and ZeroGPT as researchers' most frequently used tools, followed by less frequent ones, such as the TF-IDF method. These results are related to the work (Kutbi et al., 2024) Because they used the algorithm to identify the inconsistency of AI-generated text, accompanied by other methods such as random oversampling and logistic regression. Likewise, the work (Guleria et al., 2023) the GPTZero tool was used to identify that the articles generated by this model lacked real citations and references. Finally, in (Abani et al., 2023), several tools were used, such as a Plagiarism detector, AI Text Classifier and the most famous and repeated software in the

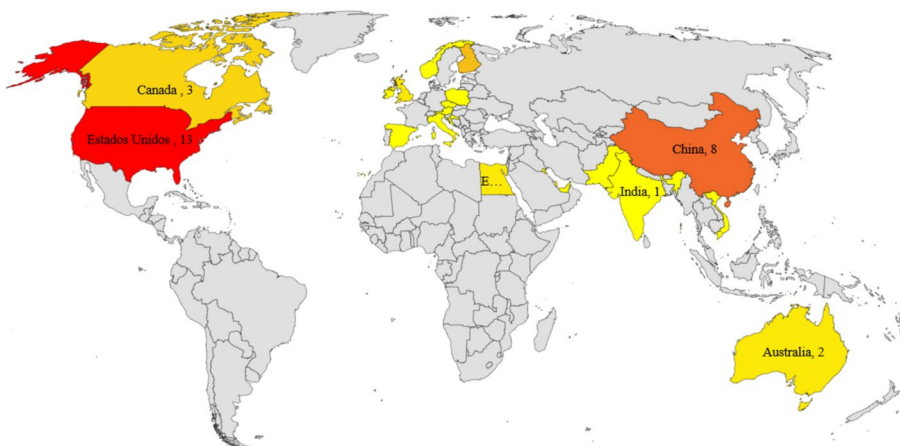


Fig. 6 Publications by country

studies of our results, Turnitin, thus identifying that this type of text is much more verbal and lacks creativity compared to those written by humans. Table 2 shows other tools for detecting plagiarism attempts using AI-based language models.

The results obtained offer an encouraging view on the development of tools to detect plagiarism in the context of language models. However, it is essential to recognize the limitations of these tools and to address the problem comprehensively, combining automated analysis with human evaluation and considering the ethical implications.

RQ2 What strategies are used to regulate the use of AI in writing scientific articles?

Just as there are tools to identify when a student has committed plagiarism using pre-trained language models, there are also disciplines and knowledge that go a long way to regulate the use of these tools and avoid falling into rational obsolescence. The findings around this question mostly relate to implementing measures, guidelines, policies and rules to preserve human reasoning and promoting controlled educational environments that ensure academic integrity. These findings differ slightly from those found

Table 2 Text-matching tools

#	Text-matching tools	Quantity	References
1	Turnitin	11	(Abdelhafiz et al., 2024; Eaton, 2023; Kacena et al., 2024; Khalaf, 2024; J. Q. J. Liu et al., 2024; Mamo et al., 2024; Margetts et al., 2024; Perkins et al., 2024; Steponenaite & Barakat, 2023; Stutz et al., 2023; Xie et al., 2023)
2	Algorithm diff	1	(Z. Liu et al., 2023)
3	Cosine similarity algorithm	1	(Z. Liu et al., 2023)
4	Jaccard algorithm	2	(Bašić et al., 2023; Z. Liu et al., 2023)
5	TF-IDF algorithm	1	(Z. Liu et al., 2023)
6	SimCSE algorithm	1	(Z. Liu et al., 2023)
7	SBERT algorithm	1	(Z. Liu et al., 2023)
8	Paraphrasingtool.ai	1	(Steponenaite & Barakat, 2023)
9	SmallSEOTools	1	(Steponenaite & Barakat, 2023)
10	Immovable element embedding	1	(Wang et al., 2023)
11	Q-Text	1	(Zhao et al., 2024)
12	PlagScan	1	(Bašić et al., 2023)
13	R Studio	1	(Bašić et al., 2023)
14	Writer	1	(Elkhatat et al., 2023)
15	Copyleaks	2	(Elkhatat et al., 2023; Kacena et al., 2024)
16	GPTZero	3	(Eaton, 2023; Elkhatat et al., 2023; J. Q. J. Liu et al., 2024)
17	CrossPlag	2	(Elkhatat et al., 2023; Ibrahim, 2023)
18	Originality.ai	1	(J. Q. J. Liu et al., 2024)
19	ZeroGPT	1	(J. Q. J. Liu et al., 2024)
20	Content at Scale	1	(J. Q. J. Liu et al., 2024)
21	GPT-2 Output Detector	2	(Ibrahim, 2023; J. Q. J. Liu et al., 2024)
22	Grammarly	1	(Safrai & Orwig, 2024)
23	Quetext	1	(Safrai & Orwig, 2024)
24	iThenticate	1	(Khalil & Er, 2023)

in the paper (E. Jo & Kim, 2023), which indicates that AI can be very useful in writing and expressing ideas but that students should still be knowledgeable about the subject. However, (Črček & Patekar, 2023) is very much related to providing clear guidelines to preserve human reasoning and promote the ethical use of chatbots. Similarly, (Singh, 2023) argues the need for responsible use by students and implementing environments that encourage academic integrity and appropriate use of these tools. On the other hand, digital literacy, specifically AI, is a practice highlighted by the study (Titko et al., 2023) to train the skills of staff and teams in academic areas in this field, as well as the need for periodic reviews by external agents and ethical analysis of the text generated. Table 3 presents the most recurrent strategies identified after the review of the studies.

It is clear that the educational and research community is aware of the challenges and opportunities posed by artificial intelligence in text production, which is why the results presented underline the importance of proactively and collaboratively addressing the regulation of the use of linguistic models in academia. There is a need to develop clear regulatory frameworks, promote digital literacy and foster a culture of academic integrity.

RQ3 What techniques are used to detect AI-generated text?

While the tools outlined above can detect attempted plagiarism by those who use AI, there are also techniques, methods and software that can detect when a text has synthetically generated fragments. These techniques and methods can vary greatly depending on their application. There is software such as ZeroGPT or DetectGPT, more analytical methods such as measuring perplexity and explosiveness, and even more traditional techniques such as asking the AI itself whether it recognizes the generated text or correctly cites the information used.

The methods presented are related to the study (Ring, 2023), which identifies the lack of AI's ability to correctly cite and reference generated data, encouraging editors not to neglect this subtle but necessary practice. Furthermore, the study (Lund et al., 2023) also favors these practices, acknowledging that AI-based chatbots do not have the skills to cite the references in the generated text. On the other hand, the results of (Balel, 2023) are closely related to the technique of recognizing content that has been previously published, or that already exists in other repositories. There is uncertainty about what information ChatGPT will use to generate the answers. Table 4 shows other techniques.

The results presented provide a detailed and up-to-date overview of the various techniques and tools available for detecting AI-generated texts, such as those produced by ChatGPT. It is clear that the scientific and technological community is investing significant effort in developing increasingly sophisticated methods to identify this type of content.

Undoubtedly, the use of AI in writing has been increasing over the years, and in turn, tools and techniques to identify when a text was generated by these tools, as well as strategies and measures to focus its use towards a more balanced one and implement it efficiently in different fields, have emerged.

As many text-matching tools that help the identification of plagiarism detection could arise with the passage of time, future research could focus on improving detection tools by developing more sophisticated algorithms that analyze multiple text features, while also establishing evaluation standards through larger and diverse datasets, clear metrics, and systematic tool comparisons. Additionally, addressing ethical and social implications by creating ethical frameworks, studying the impact on education and research, and fostering interdisciplinary collaboration is essential. Lastly, research should anticipate the evolution

Table 3 Strategies to regulate the use of AI

#	Strategies to regulate the use of AI	Quantity	References
1	Measures to preserve human reasoning	21	(Abdelhafiz et al., 2024; Akbar, Khan, & Liang, 2023; Akbar, Khan, Mahmood, et al., 2023; Awosanya et al., 2024; Buckingham Shum & Luckin, 2019; Chan, 2023; Chan & Lee, 2023; Cole et al., 2022; Elkhatat, 2023; Fontenelle-Tereshchuk, 2024; Huang et al., 2023; Ibrahim, 2023; Khalaf, 2024; Lewis et al., 2020; Michel-Villareal et al., 2023; Nam & Bai, 2023; Scott & van der Poel, 2024; Sedlbauer et al., 2024; Stutz et al., 2023; Weidmann, 2024; Xie et al., 2023)
2	AI Literacy	13	(Abdelhafiz et al., 2024; Akbar, Khan, & Liang, 2023; Awosanya et al., 2024; Black et al., 2024; Chan & Hu, 2023; Chan & Lee, 2023; Davy Tsz Kit et al., 2022; Hadley, 2022; Huang et al., 2023; Lewis et al., 2020; Sedlbauer et al., 2024; Strzelecki, 2024; Wang et al., 2023)
3	Combining Cultural-Historical and Meta-Design theory	1	(Lundin et al., 2023)
4	Ethical analysis of algorithms, people and data	2	(Astobiza et al., 2021; Chan & Hu, 2023)
5	Moderation tools	5	(Hadley, 2022; Hayawi et al., 2024; Kooli, 2023; Sakib et al., 2023; Xie et al., 2023)
6	Motivational environments and academic integrity	13	(Akbar, Khan, & Liang, 2023; Awosanya et al., 2024; Chan & Hu, 2023; Elkhatat, 2023; Fontenelle-Tereshchuk, 2024; Khalaf, 2024; Kooli, 2023; Mamo et al., 2024; Michel-Villareal et al., 2023; Sedlbauer et al., 2024; Steponaite & Barakat, 2023; Stutz et al., 2023; Xie et al., 2023)
7	Third-party review	5	(Awosanya et al., 2024; Celik, 2023; Fontenelle-Tereshchuk, 2024; Hadley, 2022; Ibrahim, 2023)
8	Logistic regression	1	(Hayawi et al., 2024)
9	Reducing written assignments	2	(Khalaf, 2024; Perkins et al., 2024)

Table 4 Techniques for detecting AI-generated text

#	What techniques are used to detect AI-generated text?	Quantity	References
1	Measurement of perplexity	3	(J. Q. J. Liu et al., 2024; Steponenaite & Barakat, 2023; Xie et al., 2023)
2	Explosivity measurement	2	(Steponenaite & Barakat, 2023; Xie et al., 2023)
3	GPTZero	1	(Xie et al., 2023)
4	ZeroGPT	1	(Stutz et al., 2023)
5	Cosine algorithm	1	(Raman et al., 2024)
6	DetectGPT	1	(Hoq et al., 2024)
7	RoBERTa detector	1	(Desaire et al., 2023)
8	GPTSniffer	1	(Nguyen et al., 2024)
9	Text Neutral Interest	1	(Cukurova et al., 2019)
10	Similar content in databases	2	(Cope et al., 2021; Nam & Bai, 2023)
11	Ask who wrote the text	2	(Khalil & Er, 2023; Leander & Burris, 2020)
12	Text mining	1	(Manzoor et al., 2023)
13	Quality checks	1	(Nam & Bai, 2023)
14	Feature-based approaches and deep learning	1	(Elkhatat, 2023)
15	Correct citation	1	(Eaton, 2023)
16	Simple, zero-trigger and fine-tuning classifiers	1	(Ibrahim, 2023)

of language models by exploring techniques to evade detection and working toward more transparent models.

Conclusion

After an exhaustive systematic review of 54 articles related to AI's impact on research ethics, we conclude that, although this type of technological solution offers a wide range of possibilities, there is a fine line between ethics and the abuse of its capabilities.

As evidenced in the analysis of articles, most authors provide strategies for regulating the use of AI-based language models to avoid falling into what we call "rational obsolescence", where human reasoning is no longer needed when they identify that a tool can automate these tasks and complete them in significantly less time.

It is also recommended that educators apply practical approaches that include the integration of the tools identified in this review and AI literacy, foster collaboration between educator and AI experts to design more effective pedagogical strategies, promote the use of controlled environments that allow students to use AI safely, and continuously train educators in the ethical use of technology, all with the goal of ensuring responsible teaching aligned with educational values..

Finally, we recommend further research and the development of more text-matching tools to aboard the improvement of plagiarism detection and encourage rather than attempting to mitigate or reduce the use of AI in any activity involving text writing. The results of this research promote possible related work on subsequent language models with increased

processing power, such as the more recent GPT-4o and the future GPT-5, both developed by OpenAI and in the focus of many institutions and researchers.

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